

MAT 21C Discussion 2

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§I Warm-Up

(Exercise) We will construct squares on top of the unit interval $[0, 1]$. First, we construct a square using the middle third $[1/3, 2/3]$ as the base; this creates two gaps on the left and right. On each gap, we repeat the same process: construct a square using the middle third of the gap as the base. We keep repeating this process indefinitely.

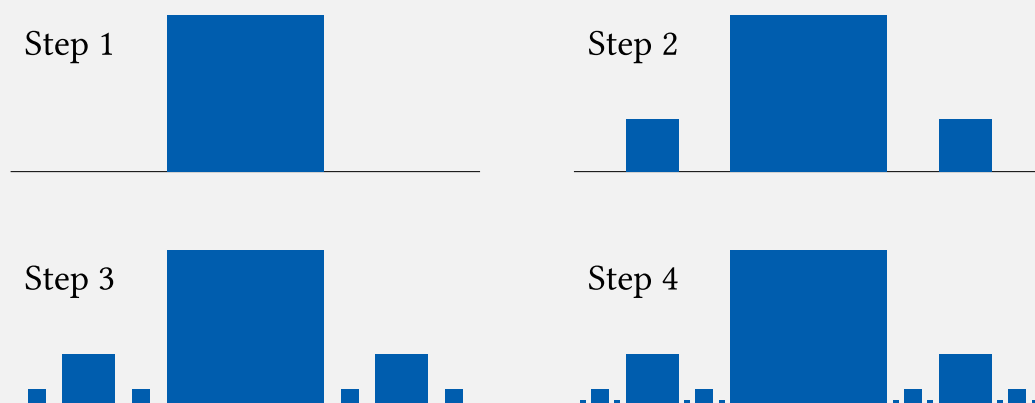


Figure 1: The first four steps of the construction.



Figure 2: The limit of the construction.

Discuss with your group how you would approach the question of how much area is covered by the squares in the limit. Can you do it with simple algebra without using a series?

Solution. Let the total area covered by the squares in the limit be A . The entire construction can be seen as two scaled copies of itself (the left and right halves) plus the first square (the middle one). Suppose that each copy is scaled by a factor of r , then we have the equation

$$A - 2rA = \left(\frac{1}{3}\right)^2.$$

Solving for A , we get

$$A = \frac{(1/3)^2}{1 - 2r}.$$

The factor r can be determined by taking the area of a square added in Step 2 (which has width $1/9$) and dividing it by the area of the square added in Step 1 (which has width $1/3$), which gives us

$$r = \frac{(1/9)^2}{(1/3)^2} = \frac{1}{9}.$$

Therefore, the total area covered by the squares in the limit is

$$A = \frac{1/9}{1 - 2 \cdot (1/9)} = \frac{1}{9 - 2} = \frac{1}{7}.$$

§II Geometric Series

The formula for geometric series is a special case of the trick we saw in the warm-up: we can get a closed form for a sum by subtracting a scaled version of itself.

If we are looking at the sum of six terms

$$s_6 = a_1 + a_2 + a_3 + a_4 + a_5 + a_6,$$

with common ratio r , we can compare it with the scaled version

$$rs_6 = a_2 + a_3 + a_4 + a_5 + a_6 + ra_6$$

and get the equation

$$s_6 - rs_6 = a_1 - ra_6.$$

$$s_6 = a_1 + a_2 + a_3 + a_4 + a_5 + a_6$$

$$rs_6 = a_2 + a_3 + a_4 + a_5 + a_6 + ra_6$$

$$s_6 - rs_6 = a_1 - ra_6$$

Figure 3: The scaling trick applied to a partial geometric series.

Therefore, we can solve for s_6 and get

$$s_6 = a_1 + a_2 + a_3 + a_4 + a_5 + a_6 = \frac{a_1 - ra_6}{1 - r}.$$

This works very generally: the sum of N terms differing by a common ratio r is given by

$$a_1 + a_2 + \cdots + a_N = \frac{a_1 - ra_N}{1 - r},$$

or in words

$$\text{the sum of terms with common ratio } r = \frac{(\text{the first term}) - r \cdot (\text{the last term})}{1 - r}.$$

A **geometric series** is a series

$$a_1 + a_2 + a_3 + \cdots$$

whose terms are related by a common ratio r :

$$a_1 \xrightarrow{\times r} a_2 \xrightarrow{\times r} a_3 \xrightarrow{\times r} a_4 \xrightarrow{\times r} a_5 \xrightarrow{\times r} \cdots.$$

Taking the limit of $a_1 + a_2 + \cdots + a_N$ as N goes to infinity (and applying the n -th term test when $\lim_{N \rightarrow \infty} a_N \neq 0$), we get the formula for the geometric series

$$a_1 + a_2 + a_3 + \cdots = \begin{cases} \frac{a_1}{1-r} & \text{if } \lim_{N \rightarrow \infty} a_N = 0, \\ \text{diverges} & \text{otherwise.} \end{cases}$$

We have $\lim_{N \rightarrow \infty} a_N = 0$ if and only if $|r| < 1$ or $a_1 = 0$, so we can also write the formula as

$$a_1 + a_2 + a_3 + \cdots = \begin{cases} \frac{a_1}{1-r} & \text{if } |r| < 1 \text{ or } a_1 = 0, \\ \text{diverges} & \text{if } |r| \geq 1 \text{ and } a_1 \neq 0. \end{cases}$$

(Exercise) Determine whether the following series converge, if they do, find their sums.

1. $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \cdots$

2. $\sum_{n=2}^{\infty} \frac{(-2)^n}{3^{n+1}}$

3. $\sum_{n=1}^{\infty} 3^{(-2n)}$

4. (Bonus) Come up with a method using geometric series to count all the natural numbers in under 1 second!

Solution.

1. This is a geometric series with ratio $1/2$, so it converges and its sum is the first term divided by $1 - 1/2$, which is

$$\frac{1/2}{1 - 1/2} = 1.$$

2. As we increase n by 1, we see that the terms are multiplied by $\frac{-2}{3}$, so this is a geometric series with ratio $-2/3$. Since $|-2/3| < 1$, the series converges and its sum is the first term divided by $1 - (-2/3)$, which is

$$\frac{(-2)^2/3^3}{1 - (-2/3)} = \frac{4/27}{5/3} = \frac{4}{45}.$$

3. Recall that

$$3^{-2n} = \left(\frac{1}{3}\right)^{2n} = \left(\frac{1}{3^2}\right)^n,$$

so this is a geometric series with ratio $1/9$. Since $|1/9| < 1$, the series converges and its sum is the first term divided by $1 - 1/9$, which is

$$\frac{1/9}{1 - 1/9} = \frac{1}{8}.$$

4. There are many ways to do this, one of them is to use part 1. When we start our timer, we count 1 after $1/2$ second, then we count 2 after another $1/4$ second, then we count 3 after another $1/8$ second, and so on. Every natural number will be counted sometime before our timer reaches 1 second!

§III Integral Test and Direct Comparison Test

Both the integral test and the direct comparison test are based on the following principle.

(Principle of nonnegative series) If the series

$$a_1 + a_2 + a_3 + a_4 + \cdots$$

consists only of nonnegative terms (that is $a_n \geq 0$ for all n), then it either converges to a finite nonnegative number or diverges to infinity. The limit is always well-defined!

This is due to the monotone convergence theorem and you can think about how they relate to each other. The principle is not necessarily true for series with negative terms, for example $1 - 1 + 1 - 1 + \cdots$ does not have a well-defined limit.

We have a simple characterization of series with nonnegative terms: it converges when there is a finite upper bound, and it diverges when it is infinite (we do not have to worry about divergence in the sense of having ill-defined limit like $1 - 1 + 1 - 1 + \cdots$).

(Integral test) Let $a_N, a_{N+1}, a_{N+2}, \dots$ be a sequence of nonnegative terms. Suppose that $a_n = f(n)$, where $f(x)$ is a nonnegative and **decreasing** function of x for all $x \geq N$. Then, we have the inequality

$$\int_N^\infty f(x) dx \leq \sum_{n=N}^\infty a_n \leq a_N + \int_N^\infty f(x) dx$$

due to the following figures.

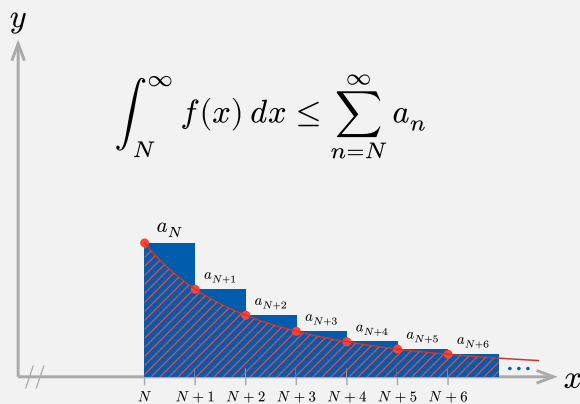


Figure 4: Lower bound for the series. Notice that we are swinging the rectangles to the right to stay above the curve.

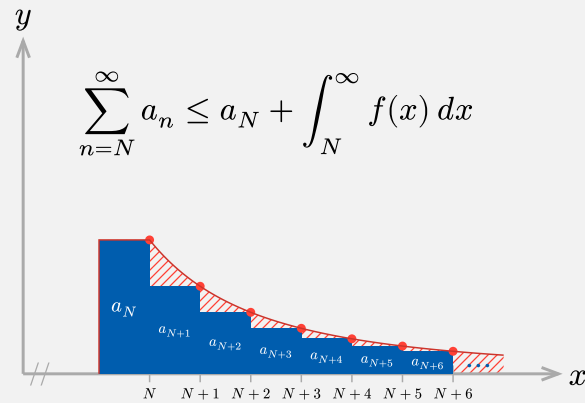


Figure 5: Upper bound for the series. Notice that we are swinging the rectangles to the left to stay under the curve.

In particular,

- if $\int_N^{\infty} f(x) dx = L$ is finite, then $\sum_{n=N}^{\infty} a_n \leq a_N + L$ is also finite by the upper bound;
- if $\int_N^{\infty} f(x) dx = \infty$, then $\sum_{n=N}^{\infty} a_n = \infty$ by the lower bound.

It is important that you show that the function $f(x)$ is decreasing when it is not obvious. For example, it is not obvious that $f(x) = \frac{\ln x}{x}$ is decreasing for $x \geq 3$ because both the numerator and denominator are increasing. To show that $f(x)$ is decreasing, we need to compute its derivative and show that it is negative for $x \geq 3$:

$$f'(x) = \frac{x^{\frac{1}{x}} - \ln x}{x^2} = \frac{1 - \ln x}{x^2} < 0 \quad \text{for } x \geq 3 \text{ since } \ln x \geq \ln 3 > 1.$$

The direct comparison test is an intuitive result.

(Direct comparison test) Let $\sum_{n=N}^{\infty} a_n$ and $\sum_{n=N}^{\infty} b_n$ be two series with $0 \leq a_n \leq b_n$. Then,

$$0 \leq \sum_{n=N}^{\infty} a_n \leq \sum_{n=N}^{\infty} b_n.$$

In particular,

- if $\sum_{n=N}^{\infty} b_n$ is finite, then $\sum_{n=N}^{\infty} a_n$ is also finite by the upper bound;
- if $\sum_{n=N}^{\infty} a_n = \infty$, then $\sum_{n=N}^{\infty} b_n = \infty$ by the lower bound.

The comparison test is often used in conjunction with the integral test: we compare a complicated series with a simpler one that is either geometric or yields a simple integral when we apply the integral test.

(Exercise) Determine whether the following series converge or diverge. If they converge, provide a lower bound and an upper bound.

$$1. \ 1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \frac{1}{25} + \dots$$

(Bonus: can you find a pair of bounds whose difference is less than 0.05?)

$$2. \ \sum_{n=1}^{\infty} \frac{2 + \cos n}{n}$$

$$3. \ \sum_{n=1}^{\infty} \frac{2 + \cos n}{n^2}$$

$$4. \ \sum_{n=1}^{\infty} \frac{1}{(n-4)^2 + 1} \text{ (hint: } \int_0^{\infty} \frac{1}{x^2 + 1} dx = \frac{\pi}{2} \text{)}$$

Solution.

1. This series can be written as

$$\sum_{n=1}^{\infty} \frac{1}{n^2}.$$

Notice that $\frac{1}{x^2}$ is nonnegative and decreasing for $x \geq 1$, so we can apply the integral test and write

$$\int_1^{\infty} \frac{1}{x^2} dx \leq \sum_{n=1}^{\infty} \frac{1}{n^2} \leq 1 + \int_1^{\infty} \frac{1}{x^2} dx.$$

We calculate that

$$\int_1^{\infty} \frac{1}{x^2} dx = \left[-\frac{1}{x} \right]_1^{\infty} = 1,$$

so the series converges and its sum is between 1 and 2.

We can get better and better bounds by extracting initial terms and applying the integral test to the rest of the series, for example, we can break the series into

$$1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \frac{1}{25} + \dots = 1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \sum_{n=5}^{\infty} \frac{1}{n^2}.$$

Applying the integral test to the last part, we have

$$\int_5^{\infty} \frac{1}{x^2} dx \leq \sum_{n=5}^{\infty} \frac{1}{n^2} \leq \frac{1}{25} + \int_5^{\infty} \frac{1}{x^2} dx.$$

Calculating the integral, we have

$$\int_5^{\infty} \frac{1}{x^2} dx = \left[-\frac{1}{x} \right]_5^{\infty} = \frac{1}{5},$$

thus, the original series is between

$$1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \frac{1}{5} \quad \text{and} \quad 1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \frac{1}{25} + \frac{1}{5}.$$

In decimals, we find that the sum is between 1.623 and 1.664.

2. Notice that this series is nonnegative because $2 + \cos n \geq 1 \geq 0$, so we can apply the direct comparison test. Because $1 \leq 2 + \cos n \leq 3$, we have

$$\frac{1}{n} \leq \frac{2 + \cos n}{n} \leq \frac{3}{n}.$$

Since $\sum_{n=1}^{\infty} \frac{1}{n} = \infty$ by the integral test:

$$\int_1^{\infty} \frac{1}{x} dx = [\ln x]_1^{\infty} = \infty,$$

we also have that $\sum_{n=1}^{\infty} \frac{2 + \cos n}{n} = \infty$ diverges.

3. Still using $1 \leq 2 + \cos n \leq 3$, we have

$$\frac{1}{n^2} \leq \frac{2 + \cos n}{n^2} \leq \frac{3}{n^2}.$$

Thus, by part 1, we have

$$1 \leq \sum_{n=1}^{\infty} \frac{1}{n^2} \leq \sum_{n=1}^{\infty} \frac{2 + \cos n}{n^2} \leq 3 \sum_{n=1}^{\infty} \frac{1}{n^2} \leq 6.$$

4. It is very tempting to apply the integral test since the hint literally tells us how to compute the integral, but notice that the terms in

$$\sum_{n=1}^{\infty} \frac{1}{(n-4)^2 + 1} = \frac{1}{9+1} + \frac{1}{4+1} + \frac{1}{1+1} + \frac{1}{0+1} + \frac{1}{1+1} + \frac{1}{4+1} + \frac{1}{9+1} + \dots$$

are not decreasing, so any interpolating function would also not be decreasing. The good news is that we can extract the first three terms

$$\frac{1}{9+1} + \frac{1}{4+1} + \frac{1}{1+1} = \frac{1}{10} + \frac{1}{5} + \frac{1}{2} = \frac{8}{10},$$

and apply the integral test to the rest of the series

$$\frac{1}{0+1} + \frac{1}{1+1} + \frac{1}{4+1} + \frac{1}{9+1} + \dots = \sum_{n=0}^{\infty} \frac{1}{n^2 + 1}.$$

Since $\frac{1}{x^2+1}$ is nonnegative and decreasing for $x \geq 0$, we have

$$\int_0^{\infty} \frac{1}{x^2 + 1} dx \leq \sum_{n=0}^{\infty} \frac{1}{n^2 + 1} \leq \frac{1}{0+1} + \int_0^{\infty} \frac{1}{x^2 + 1} dx.$$

Using the hint, we have

$$\frac{\pi}{2} \leq \sum_{n=0}^{\infty} \frac{1}{n^2 + 1} \leq 1 + \frac{\pi}{2}.$$

Therefore, the original series converges and its sum is between

$$\frac{8}{10} + \frac{\pi}{2} \quad \text{and} \quad \frac{8}{10} + 1 + \frac{\pi}{2}.$$